

Senior Data Analytics Portfolio

The Skills Big Employers Test — Flagship Project Blueprint & 6-Month Roadmap

Prepared for Rashadul Islam Roman · Toronto · September 21, 2026 · Rev 6 — the three-audience edition: hiring managers, PhD professors, *and now business clients* · adds the client-facing offer (Section 14) that re-reads the NorthLedger build as a done-for-you analytics service — messy-data intake, governed pipelines, live dashboards, automated daily insight — plus the portfolio-website packaging row · prior revs: Rigor Standard + manager playbook (Rev 5), Legacy-to-Modern map (Rev 3), finance & economics layer (Rev 4)

A complete, evidence-based plan to build ONE portfolio project that proves senior-level data analytics, data science and business-intelligence skill to Big Tech (Amazon, Meta, Google class) and banks and financial firms (RBC, TD, CIBC, JPMorgan class). Every requirement below is grounded in real 2025–2026 postings and hiring research — the full evidence reports live in [~/data-analytics-portfolio/research/](#).

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Executive Summary

The gap between a good candidate and a senior interview is not "knows SQL and Python" — everyone applies with that. It is end-to-end proof, in one artifact, that you can take raw, messy, large data to a decision, and defend how and why. A hiring manager who has screened 1,000+ portfolios estimates that roughly 95% of projects fail to connect the work to a decision or outcome [48], and reviewers give a portfolio 30 seconds to 3 minutes before deciding whether to keep reading [49]. The candidates who win build evidence, not notebooks.

The research behind this blueprint confirms the bar from both directions. SQL is named in 52.9% of data-analyst postings market-wide [28] and in 100% of the bank and financial postings sampled for this report; Python in 31.2% of analyst postings [28] and 57% of data-scientist postings [29]. Power BI appears in 29% of analyst postings [28] and is the Canadian-bank default — a live TD posting lists "Power BI certification" as an asset [34]. Every senior Big-Tech loop tests four things: a hard SQL screen, an experimentation/statistics round, a product-metrics case, and communication [3][8][21]. And generative-AI skill now appears explicitly in 6.9% of analyst postings — at roughly a \$22K salary premium [30].

Rev 2 extends the evidence beyond that first sweep: the full North-American market (33 employers across insurance, healthcare payers, retail, telecom, consulting and industrial) and the EU (8 countries, ~25 ads from Berlin to Dublin). It changed the plan in four concrete ways. **Excel / Power Query became a P0 module** — it appears in 41–75% of analyst ads [75][76]. **Tableau gets a mandatory mirror dashboard** — it edges Power BI in US datasets [75]. **Airflow replaced Dagster** as the orchestration headline (231 vs 23 UK postings [84][85]). And the governance pack gained a **privacy/regulatory annex** (GDPR, EU AI Act, HIPAA/SOX context) [90][92]. Rev 3 adds **The Legacy-to-Modern Map** (Section 4): every retiring skill — SAS, Hadoop, SSIS, QlikView, VBA-era Excel reporting — paired with its modern replacement and the play for packaging the migration. Rev 4 adds the **finance & economics capability layer** (Section 2): verified against 27 finance-industry and 55 economic-data postings — AML / financial-crime analytics, IFRS 9 / ECL staging and stress-test framing, model-validation governance (SR 26-2 vocabulary), Stata / R + panel econometrics, and the markets language (VaR / FRTB / security master) — with every addition sized as a thin bolt-on to the same flagship (Section 6). Rev 5 raises the bar for both audiences — a rigor standard that survives a professor and a manager (Section 5), a manager-facing artifact system, and the **open-data catalogue**: 178 live-verified public datasets across North America, Europe and global portals (Section 10), including the US crime records and Dutch cycling data you asked for.

The plan is ONE flagship project — **NorthLedger**, a financial intelligence platform — that processes 10M+ transaction records through a modern data stack and serves three decision-makers: fraud operations, customer growth, and the executive committee. The research behind this report scopes such a build at a realistic 80–150 solo hours, which fits 10–15 hours per week over six months. It runs on \$0/month infrastructure (Section 8). It packages into the artifacts that actually get read: a GitHub case study, three Power BI dashboards, a one-page executive summary, and a 90-second walkthrough video [50][51]. It is designed so that every interview loop you will face — SQL, experimentation, metrics case, communication — has a real answer inside it.

Rev 6 adds the third audience. The same build that impresses a hiring manager and survives a professor also *sells* — a small or mid-size business sitting on messy exports, spreadsheets and siloed tools is the exact problem NorthLedger's pipeline solves on synthetic data. Section 14 turns the project into a client offer: a data-audit entry offer, an automated build (clean → model → dashboard → daily insight), a lightweight monthly operate plan, and a one-page capstone that runs the whole loop end-to-end on free public data the way NorthLedger runs on transactions. The portfolio website (Section 12) now carries two doors on one page: "Hire me" and "Book a data audit" — employer and client, one build.

Sections 2–3 give the evidence and the skill-by-skill map; Section 4 maps the outdated skills to their modern replacements; Section 6 is the project itself; Sections 7–8 are the platform and stack; Section 10 is the open-data catalogue; Section 11 is the month-by-month roadmap; Section 12 is the packaging that turns the project into interview calls.

What Big Employers Actually Test

Method: live postings were read from official careers sites in September 2026 (a few via lawful mirrors where a req had closed), and interview-loop expectations come from 2025–2026 interview guides. Postings rotate fast — re-verify any specific req before citing it in an application. The point of this section is not one posting; it is the pattern across more than one hundred postings sampled in North America and the EU (Rev 2).

Big Tech: the requirements that repeat

Company (role)	What the postings literally demand	Loop notes
Amazon (BIE / Sr BIE)	"5+ years of SQL"; multi-TB datasets; "theory and practice of design of experiments"; scalable data models, semantic layers, source-of-truth datasets; Tableau/QuickSight dashboards; executive storytelling; "leveraging generative AI tools... is highly valued" [1][2]	Take-home SQL; loop tests SQL, data modeling/ETL design, metric definition, Leadership-Principle behavioral [3][4]
Meta (DS, Analytics)	6+ yrs (4+ with PhD); SQL + Python/R; experimentation design & analysis; metric definition and instrumentation; business acumen rated "expert" — above statistics [5][6][7]	Live SQL screen, stats/A-B round, product-sense rounds, behavioral; writing ≈ 18% of the workweek [7][8][9]
Google (Product / Research DS)	3+ yrs (5 preferred); SQL with BigQuery idioms, statistics, A/B testing, causal inference; Python/R; "end-to-end analysis... including requirements specification... and presentations" [11][12]	SQL round, stats/causal round, ML round, product-analytics case [13]
Microsoft (Senior DS)	SQL + KQL/Kusto at scale; A/B testing and offline evaluation; Python/R; Azure stack — ADX/Kusto, Synapse, Fabric, Databricks [14][15]	SQL/KQL on large datasets, telemetry analysis, ML depth
Apple (DS / Senior DS)	"Expert SQL"; Python (pandas/NumPy/sklearn/PySpark); experiment design incl. power; quasi-experimental methods — "diff-in-diff, propensity matching, synthetic controls"; Tableau; "turning analysis and models into shipped decisions with measurable impact" [16][17][18]	Screens test power analysis, causal inference, visualization, communication [16]
Netflix (DS 5)	Experimentation + causal inference "beyond vanilla A/B tests"; treatment-effect estimation; SQL + Python incl. dashboards and automation; "define and evolve... measurement frameworks" [19][20]	Experimentation-heavy loop; experiment readouts are the core exercise
Uber (DS / Scientist)	Advanced SQL; A/B testing and statistical methods; causal inference; Python/R at scale; product sense skewed to fraud/risk/marketplace economics [21]	Dedicated SQL and data modeling rounds; business cases — under-preparing them is the #1 observed failure [21]
Airbnb (Senior DS)	5+ yrs; SQL at scale + Python/R; experimentation + causal observational analysis; ML for marketplace settings; translating methodology to non-technical audiences [22][23]	Product sense & metrics + experimentation ≈ 40% of the evaluation
Spotify (Senior DS)	A/B testing and experimental design "required"; causal inference incl. launches; Bayesian and regression modelling; strong SQL + advanced R/Python; "extraordinary storyteller and communicator" [24]	Communication weighted heavily
NVIDIA (Senior DS)	Production-grade Python; probability/statistics + ML; optimization/OR (linear programming, network flow, bandits, OR-Tools); Spark/Databricks/Delta Lake; MLflow deployment; "leverage agentic AI" [25]	Python quality, ML + optimization depth, forecasting, deployment

The pattern across all ten: every senior loop contains (1) a hard SQL screen, (2) an experimentation/statistics round, (3) a product-metrics or business case, and (4) a communication/behavioral round. ML depth differentiates only in some companies (Netflix, NVIDIA, Airbnb, Google Research); take-homes appear mainly at Amazon and sometimes Meta. Market-wide posting baselines: SQL 52.9%, Python 31.2%, R 24.9%, Power BI 29.0%, Tableau 26.2% of analyst postings [28]; for data-scientist postings: Python 57%, machine learning 69%, R 33%, SQL 30% [29].

What "senior" means in these postings is scope, not years: owning the measurement framework [1], defining KPIs where none exist [18], and setting experimentation standards others adopt [20]. It means turning "open-ended problems into structured analytical approaches" [10] and influencing shipped decisions. A pivoting candidate cannot manufacture six years — so the portfolio must manufacture the scope these postings describe.

Banks & financial firms: what the postings demand

SQL is required in 100% of the financial-sector postings sampled (Canadian big six plus JPMorgan, Goldman Sachs, Morgan Stanley, Capital One, Amex, Wealthsimple, Sept 2026). SAS remains an active production requirement in Canadian banks — TD is running a "SAS Grid → SAS MLE conversion" program [46] and BMO asks analysts to "execute and maintain existing SAS scripts" [39]. Power BI is the Canadian default: TD requires "advanced Power BI development" and lists certification as an asset [34], and BMO builds "executive-level Power BI dashboards" [39]. Scotiabank and CIBC list Power BI/Tableau/Power Platform [36][40]. Tableau still dominates US firms [43].

Employer	Stack signals	Senior gate vocabulary	Certs named in postings
RBC	Python/SQL/SAS, Tableau and Power BI, Snowflake; AI-agent work "nice to have" [32][33]	Regulatory & financial reporting; "exceptional storytelling... to Senior Leadership" [33]	—
TD	Advanced Power BI + Power Automate, SQL, PySpark, SAS, Azure + Databricks [34][35][46]	IFRS 9/ECL, RWA, stress testing; fraud/CIAM governance; marketing measurement [34][46]	"Power BI certification" as an asset [34]
Scotiabank	SQL, Power BI/Tableau/Excel, Snowflake/Hadoop/GCP [36][37]	AML/ATF, anomaly detection, model productionisation [37]	—
BMO	Python, SQL, SAS, Power BI, Spark/Scala, AWS/Azure/GCP [38][39]	Commercial credit risk; lending & decisioning; fraud [38][39]	—
CIBC	Power Platform (Power BI/Automate/Apps), Azure Databricks "a must", Python/R, SAS + Oracle SQL [40]	GenAI/LLM solutions; governance, risk & controls; exec reporting [40]	Degree "ideally Masters" [40]
National Bank	SAS, Python, SQL; IFRS 9 and stress-testing models [41]	Credit risk, RWA, Economic Capital, model governance [41]	SAS cert and CFA/FRM "would be an asset" [41]
US majors (JPMC, GS, CapOne, Amex)	Python/R/SQL, Snowflake, Alteryx, Tableau; Power BI at Amex/GS [42][43][45]	Data governance (metadata, lineage, CDEs); attribution; pricing analytics [42] [43][45]	Six Sigma preferred at CapOne [42]
Fintech benchmark (Wealthsimple)	dbt + Airflow, data-mart architecture, AWS S3/Redshift + Snowflake, Git/PRs, Python [44]	Analytics engineering; profitability & regulatory reporting [44]	—

Toronto specifics: "senior" typically means 5+ years [33][38]; hybrid in-office 3 days/week is common; observed salary bands include TD \$96.9K–\$136.8K [34], BMO Sr. Data Analyst \$75.9K–\$141.9K [38], Wealthsimple \$151K–\$189K for senior analytics engineering [44]. The seniority tiebreaker is domain fluency — postings gate on risk/regulatory vocabulary (IFRS 9, ECL, RWA, AML/ATF, credit lifecycle) [41][46][37] more than on additional tools.

The wider North-American market: insurance, healthcare, retail, telecom, consulting

Beyond tech and banks, 33 employers were sampled across insurance, healthcare payers, retail, telecom, consulting and industrial (US + Canada, Sept 2026). What a tech-and-banking-only sweep misses:

- **The tool spine changes:** SQL + Power BI (DAX, Power Query, RLS/OLS) + Snowflake + Azure/Databricks + SAS + **advanced Excel/VBA** is the norm — Optum accepts "SQL and at least one additional analytics or scripting tool such as Python, R, or advanced Excel" [117]; Manulife's analyst ad wants "MS Excel, VBA, Power Platforms" plus "4+ years building advanced Power BI solutions (data modeling, complex DAX, Power Query, RLS/OLS)" [122]; Loblaw lists "Python and VBA scripts to automate" over BigQuery/Teradata [123].
- **SAS is a live requirement, not nostalgia:** Elevance wants "SAS or equivalents... Teradata, SQL" [118]; Progressive runs Snowflake/SQL/R/Python with SAS in its stack [119]; KPMG asks for "Python, R, Alteryx, SQL, SAS" [120].
- **Alteryx and KNIME are named tools** in retail, consulting and audit analytics — Walmart (SQL, Tableau, "Knime, Alteryx, R, Tableau Prep") [121], KPMG [120], and Deloitte's flagship req open across 32 locations [130]. Canadian Tire's Senior Data Analyst asks for "particularly KNIME" [124].
- **Microsoft Fabric and Copilot are in the duties:** Allstate's data-engineering req is "Microsoft Fabric / Financial Data" [128]; Manulife: "Using GenAI-based assistants to enhance SQL, Python, and Power BI development efficiency" [122].

- **A legacy BI estate matters at some shops:** Teradata, DB2/BTEQ, SAP BW/HANA, MicroStrategy, Adobe Analytics, Qlik, Domo, KNIME appear in live ads [125][126]. Name them only if a target shop runs them — Anaplan never appeared in this sample, and PCI-DSS is a payments vocabulary, not a non-tech one.
- **Domain vocabulary gates the senior band:** HIPAA and CMS Star Ratings / HEDIS / CAHPS in health [117][129]. Reinsurance and catastrophe modelling gate P&C roles; TMS/supply-chain optimisation and SOX/audit controls gate retail and audit roles [121][131].
- **Work-model reality is harsher than tech:** AT&T requires "a minimum of 5 days per week" in office [127]; Loblaw and Canadian Tire run 4 days [123][124]; and Canadian non-tech senior pay lags sharply (Loblaw \$60–82.5K CAD vs CVS Principal \$144–288K USD) — a US-remote application strategy pays more at equal skill. Canadian insurance stacks mirror the pattern (Sun Life runs "Advanced SQL Server / AWS Athena" [132]).

The EU: what changes across the Atlantic

Eight countries sampled — Germany, UK, Netherlands, France, Nordics, Ireland, Spain, Switzerland — with ~25 ads from SAP, Siemens, Zalando, Delivery Hero, Deutsche Bank, Wise, Barclays, HSBC, Revolut, ING, Adyen, Booking.com, BNP Paribas, Société Générale, Nordea, Klarna, Spotify, Novo Nordisk, AIB, Bank of Ireland, Santander, BBVA and Swiss Re.

- **The floor is identical:** SQL in 100% of the sampled ads; Python the second language; and experimentation/statistics are *more explicit than in NA analyst ads* — Booking wants "experimental design, A/B testing, statistical inference" and Zalando "causal inference" [103][115]. Bank of Ireland asks for "at least one of SQL, Tableau, Python, R, SAS" [110], and Santander runs LTV/CAC attribution models [111].
- **BI mix flips by country:** Power BI dominates Germany, Netherlands, Ireland, Spain and Switzerland [107][109], and HSBC names Power BI, Tableau and QlikView side by side [105]. Tableau's UK demand nearly tripled year-over-year — 1,581 postings vs Power BI's 2,644 in the six months to mid-2026 [80][88]. Looker shows at tech firms [106], while QlikSense is still live in the Nordics [114].
- **Vendor-native stacks:** SAP Datasphere/HANA and Business Data Cloud [101]. Microsoft Fabric shows at AIB ("SQL, Python and Fabric") [109]; BigQuery at Zalando and Delivery Hero [115][103]; Snowflake+Spark+dbt at Booking.
- **Regulatory literacy is a hiring criterion:** GDPR and BCBS 239 appear in EU bank ads; model-risk management, validation and "model documentation" at Nordea [108]; "explainable ML, fairness... governance" at BBVA [116]. Swiss Re's finance-analytics roles sit on IFRS 17/IFRS 9 valuation governance [112]. The EU AI Act is now an ad keyword — 24 UK postings in six months at a £107,500 median [92].
- **Language, country by country:** Nordics, Netherlands tech, Ireland and UK ads are English-only; Germany, France, Spain and Switzerland often require the local language — Santander says "Fluent in English and Spanish (Required)" [111], and roughly half of German firms require at least B2 German [93]. 55% of Berlin visa-sponsoring roles want fluent German [94].
- **AI-native expectations sit inside the requirements block:** RAG/LangGraph (ING) [102], "agentic AI... multi-agent architectures" (SAP) [101], and named AI coding tools (Novo Nordisk lists Cursor, Claude Code, GitHub Copilot) [113].
- **Practical mechanics:** salary ranges are landing in ads under the EU Pay Transparency Directive [100] — ING's ad shows €5,677–9,137/month [102]; Wise £60–85K [104]. Hybrid day-counts are published with "work-from-abroad" allowances, and Berlin runs a visible visa-sponsorship ecosystem (183 roles at 38 employers) [94]. If you ever target the EU, add a privacy/regulatory annex, a model-governance document, and a local-language line to the CV [90][91].

The finance & economics capability layer

Two adjacent job families deserve their own capability track: **financial data roles** (asset managers, insurers, capital markets, exchanges, payments) and **economic data roles** (central banks, statistical agencies, consultancies, international organizations). Both were verified against 2025–2026 postings — 27 finance-industry ads across NA/EU and 55 economic-role postings — and both sit close to the existing plan; the additions are extensions, not a pivot.

Financial data roles — the differentiator is data objects and gate vocabulary, not more tooling. The objects: security master and reference data, corporate actions, portfolio holdings and exposures [169][168]. The gate vocabulary: VaR / HVaR / IRC, FRTB, IRRBB / PFE, IRB with PD / LGD, attribution, "effective challenge" [159][160][163]. The vendor triangle is understood, not owned: Bloomberg PORT / BQL / BQuant for market data [167], Aladdin / Charles River / Murex as books of record [168], AIR / RMS / WTW in insurance [172]. Model governance is mid-rewrite — the Fed/OCC/FDIC replaced SR 11-7 with SR 26-2 on 17 April 2026: a risk-based, **explicitly non-binding** letter that, critically for any project shipping an AI layer, **excludes generative and agentic AI from scope** [171]; Canada parallels it with OSFI Guideline E-

23 (effective 1 May 2027), adding model inventory, lifecycle governance, and explainability-scaled AI expectations [193]. Validation language is written into live postings [161]. CFA / FRM / PRM appear as tiebreakers, not gates [167][159]. Python and SQL remain the hard floor [164][165]. GenAI / RAG vocabulary shows up in roughly a third of these ads — BlackRock's Aladdin org, Nasdaq's LangChain / LlamalIndex stack, Swiss Re's LLM-and-RAG pricing work [177][176].

Economic data roles — the master's is the pivot credential, and the toolchain is Stata + R + Python. Bank of Canada hires master's holders as economists; the IMF calls a master's "more competitive"; the World Bank prices it at five years' experience versus three for a PhD [178][179][180]. The PhD is a hard wall only at the IMF Economist Program and the ECB graduate programme [191][183]. Econometrics is named in every cluster, without exception [188][182], and BoC's forecasting seats want it applied to near-term projections [150]. Software: Stata, R and Python consolidated; SAS survives in official statistics and MATLAB where structural modelling lives [153][154][156]. Statistics Canada's EC program is the most accessible route in the entire map — two semester courses in economics qualify on education, and the master's is an education asset [184]. World Bank economist reqs run on household surveys and applied microeconometrics [181], and economics consultancies hire MSc-level analysts for econometric estimation in Python and R [185][186]. Writing is scored at the final gate: the IMF runs a written exam and PIIE sits a coding assessment [191][187]. AI-assisted workflow literacy and version-controlled reproducibility are the newest near-universal requirements [187][189], and the senior model-development seat at the Bank of Canada pays CAD \$111,051–\$130,649 [192]. Timing: the federal APAP campaign closes 4 November 2026, and BoC runs September and February cycles [190][178].

What this changes in the build. Four thin upgrades, all on the same flagship (Section 6): an AML / financial-crime layer, an IFRS 9 / ECL staging + stress-test extension, a regulatory annex v2 with scorecard literalness, and an econ toolchain kit (Stata / R / panel econometrics). ≈25–40 hours total; no new datasets, no new projects.

What this means for your portfolio

1. **SQL depth is the price of entry everywhere**, and the bar includes query performance and reconciliation logic, not just correctness [1][21].
2. **Data modeling and a semantic layer are named senior requirements**: "scalable data models, semantic layers, and source-of-truth datasets" [1]; "a governed lakehouse... a clear semantic layer... certified metrics" [16]; "data mart architecture, design and management" [44].
3. **Experimentation and causal reasoning appear in every senior loop** — a portfolio with zero experiment design is structurally mismatched to the 2026 bar [19][24][16].
4. **ML matters when evaluated correctly**: PR-AUC and cost trade-offs over accuracy, with explainability and an honest limitations section [26][50].
5. **Power BI is the Canadian-bank default** — lead with it (Power Query, DAX, semantic models, RLS) [34][36]. US datasets actually rank Tableau first among named BI tools (28.1% vs 24.7%), so the Tableau mirror is now mandatory, not optional [76][75].
6. **Governance deliverables are explicitly named**: metadata, lineage, business definitions, reconciliation, UAT — by RBC, Capital One, Amex and Scotiabank postings [42][45][36].
7. **A GenAI component is now a hiring signal**: "leverage agentic AI" (NVIDIA) [25] and "apply emerging AI and GenAI techniques" (Amex) [45]. Canadian banks also list GenAI capability as an asset in senior roles [40][37].
8. **Storytelling is a literal requirement**: "exceptional storytelling... to Senior Leadership" (RBC) [33], "extraordinary storyteller" (Spotify) [24].
9. **Excel is a written requirement, not a legacy skill**: it appears in 41–75% of analyst ads across markets [78][77][79] — now a P0 module with a stakeholder-facing workbook.
10. **For EU or regulated-market targets**: expect GDPR/BCBS-239/model-governance vocabulary [90][108] and occasional local-language requirements [94]. Salary ranges now appear in EU ads [100], and agentic-AI tooling sits inside the requirements block [102][113].

The Skills Master Matrix

Every row maps a skill that senior postings test to the exact place the flagship build proves it. Priority: P0 = must exist before you apply; P1 = strong differentiator; P2 = only if ahead of schedule. Where the ads still name a legacy tool — SAS, Hadoop, SSIS, QlikView, VBA-era reporting — **Section 4** maps it to the modern replacement and the exact play.

Skill block	What employers actually test	Where NorthLedger proves it	Artifact they see	Priority
Advanced SQL	Window functions, CTEs, cohorts, funnels, performance, reconciliation	40+ dbt models + a 20-query showcase file (RFM, retention, attribution, time intelligence)	Repo <code>/sql</code> + dbt docs	P0
Data modeling & warehousing	Star-schema thinking, grain decisions, SCDs, marts layers, lineage	Star schema (facts + dimensions) in BigQuery; SCD2 customer dimension; three mart layers	ERD + dbt lineage graph	P0
ETL & orchestration	Idempotent pipelines, scheduling, retries, alerting, incremental loads; Airflow is the most-named orchestrator in ads [84][85]	Daily Airflow batch + backfill DAGs; incremental models; failure alerts; pipeline health page	DAG screenshots + run history	P0
Big data handling	10M+ row comfort, Parquet, partitioning, cost-aware querying	10M+ transaction pipeline; partitioned tables; Parquet staging; documented \$ per query	Volume and cost notes in case study	P0
Cloud (warehouse + storage)	Warehouse, storage, IAM basics, cost awareness	BigQuery + Cloud Storage; service accounts; \$0 free-tier story [65]	Architecture diagram	P0
Python for data	Pandas/Polars, clean code, notebooks vs packages, automation	Ingestion framework, feature engineering, modeling code, reusable <code>src/</code> package	Repo code + tests	P0
Advanced Excel & Power Query	Excel in 41–75% of analyst ads; "advanced Excel or VBA" written into non-tech postings; Power Query is the "proficient Excel" bar [78][117][97]	Stakeholder-facing workbook: pivots, Power Query refresh, scenario model, INDEX-MATCH, reconciling to the warehouse	<code>/excel</code> module + README paragraph	P0
Git & team workflow (Agile/Jira)	Git appears in 1,535 UK postings; Agile/Jira is standard team vocabulary [81][82][83]	PR history + decision log + a one-page sprint report artifact	Repo + <code>/docs</code>	P1
Stakeholder management & requirements gathering	6–14% of US analyst postings; EU evidence ranks communication highest for business analysts [75][89]	Requirements doc + sprint report + stakeholder framing woven through the case study	<code>/docs</code> + case study	P1
Statistics & A/B experimentation	Hypothesis tests, power analysis, guardrail metrics, readout decisions	Campaign A/B incrementality analysis with power analysis, CI reporting, a ship/no-ship call [19]	Experiment readout	P0
Causal inference (quasi-experimental)	Diff-in-diff, propensity matching, synthetic controls where tests are impossible [16][19]	One natural-experiment analysis with sensitivity checks	Analysis notebook	P1
Machine learning	Feature engineering, correct metrics (PR-AUC over accuracy), validation, explainability	Fraud model (XGBoost + IsolationForest), churn/CLV (GBM + SHAP), forecasting (Prophet/ARIMA)	Model cards + evaluation notebook	P0

Skill block	What employers actually test	Where NorthLedge proves it	Artifact they see	Priority	
AML & financial-crime analytics	Typology-driven rules, anomaly detection, alert triage, false-positive economics, SAR-style case narrative	TD calls AML its "top one priority" and staffs two data-science layers on it [139][140]; live BMO / RBC transaction-monitoring data-science postings [137][138]; 176 UK AML postings, median £72.5K, +9.4% YoY [142]	AML layer on Module A: structuring / velocity / mule typology rules + anomaly detection on the same transaction stream + alert-triage view + one case narrative	/fraud module extension + dashboard	P1
Econometrics & time-series analysis (panel, DiD, VAR, nowcast)	Stata + R + Python; regression tables; policy-brief writing; reproducibility	Named in every central-bank / stats-agency / consultancy cluster reviewed [148][149] [188][155][157]; Stata and R in nearly every econ posting [153] [147][146]; polyglot Stata/R wins 2026 market analysis [158]	Econ toolchain kit: panel-FE + DiD causal upgrade (clustered SEs), small VAR / nowcast on FRED, Stata do-file + R/Quarto mirror, 2-page policy brief	/econ module	P1 (economic track)
Risk & regulatory analytics (ECL staging, stress testing, validation)	IFRS 9 Stage 1/2/3 and SICR triggers, scenario-weighted PD, stress framing, "effective challenge" validation vocabulary (post SR 26-2)	TD ECL-staging role [141]; National Bank IFRS 9 / stress models [41]; BoC financial-stability stress-test asks [151]; IFRS 9 co-occurs 100% with credit-risk / validation titles [145]; stress-testing ads +39% YoY [143]; BMO FRTB delivery [162]; validation seats at Citi / PNC [161][160]	ECL/stress extension on the credit module + validation memo + regulatory annex v2 (BCBS 239 mapped to dbt controls; DORA glossary)	Credit module + /docs	P1
Financial-market data objects (security master, corporate actions, attribution)	Reference-data discipline, point-in-time correctness, Brinson attribution, vendor-platform mapping (Bloomberg / Aladdin / AIR-RMS)	Asset managers, custodians, exchanges and wealth teams name these directly [169][168] [170][166][167][174][175]	Markets spine + investment-analytics page: adjusted/unadjusted price ledger, attribution waterfall, "BQL-style query" mapping doc; VaR-lite engine with Kupiec backtest + model card	/markets module	P2 (finance-industry track)
BI & dashboarding (Power BI + Tableau mirror)	Semantic models, DAX, interactivity, performance, design; tool portability	3 Power BI reports (Executive, Fraud Ops, Growth) + one Tableau Public mirror of the same star schema; DAX showcase; RLS demo [34][76]	Live dashboard links + screenshots	P0	
Data storytelling	Framing, narrative, executive communication, "so what" [33] [24]	Case study, weekly insight memo, video walkthrough, exec one-pager	Case study + video	P0	
Data quality & governance	Tests, monitoring, dictionary, lineage, reconciliation, CDEs [42] [45]; "data quality" appears in 2,025 UK postings [86] and "data governance" in 1,527 [87]	dbt tests + freshness checks, data dictionary, lineage, reconciliation report, decision log, plus a privacy/regulatory annex (GDPR, EU AI Act, HIPAA/SOX context) [90][92]	dbt docs + /docs governance pack	P0	
AI-augmented analytics	LLM-assisted insight generation; agentic workflows [25][45]	Insight memo service + text-to-SQL demo with an evaluation harness	Repo module + video segment	P1	
SAS-aware migration story	Banks are actively migrating SAS→Python (RBC, TD, BMO) [46][39][33]	Reimplement one public SAS workflow in Python/SQL with parity testing	/migration mini-project	P1	

Skill block	What employers actually test	Where NorthLedger proves it	Artifact they see	Priority
Methodological & statistical rigor	The professor-grade layer: leakage discipline, proper backtests (Kupiec/Christoffersen/Basel; rolling-origin), calibrated scores, identified causal claims, uncertainty everywhere [194] [195][196][198]	LEAKAGE.md + BACKTEST_REPORT.md per model + calibration sections + one-page identification memo	/rigor pack + methods paper	P0
Reproducibility & auditability	Cold-clone regeneration of every number; lockfile, seeds, one-command rebuild, data contracts, lineage hashes [206]	Makefile/one-command verify + tests/ + data contracts + LEDGER.md + DECISIONS.md	Repo contract	P0
Model-risk governance literacy	Validation vocabulary (SR 26-2 / OSFI E-23), model inventory, monitoring obligations, explainability scaled to stakes [171][193]	MODEL_REGISTRY + VALIDATION in second-line voice + MONITORING + CHANGELOG	/docs governance pack	P1
Streaming / near-real-time	Event-time vs processing-time, windows (rarely hands-on for analysts)	Synthetic stream feeder + micro-batch demo; framed as extension	/streaming module	P2
Cost & ops awareness	"\$ per query", monitoring, on-call thinking (an interview talking point — no ad-keyword weight)	BigQuery cost notes, pipeline monitoring page, alerting	Case-study section + interview talking point	P1

The Legacy-to-Modern Map: Outdated Skills and Their Replacements

Every senior market carries a legacy layer — the tools that built the last decade's data estates and still appear in today's ads, mostly at banks, insurers and enterprises mid-migration. The wrong move is to treat them as either "must-learn" or "ignore". The right move is the pairing: **recognize the legacy vocabulary, build the modern replacement, and package the migration as a story**. This is the dictionary — what the market is retiring, what replaced it, and what to actually do.

- **The ads still name them.** Canadian banks run SAS in production today [46][39]. HSBC lists QlikView beside Power BI and Tableau [105], and insurance estates still sit on Teradata [118][123].
- **The interviewers are themselves mid-migration.** TD's own posting is a migration program — "SAS Grid → SAS MLE conversion" [46]. A candidate who can say "I reimplemented a legacy workflow in Python with parity tests" speaks their language before they finish explaining it.
- **Judgment is the senior signal.** Knowing what's obsolete, why, and what replaced it — and refusing to collect dead tools — is exactly the discrimination senior screens look for. You list *legacy awareness* without ever claiming obsolete mastery.

Retiring skill (don't build new)	Modern replacement (build this)	Where it still shows up	Your move
SAS as a primary tool	Python (pandas / statsmodels) + SQL-first workflows	TD's "SAS Grid → SAS MLE conversion" program [46]; BMO "execute and maintain existing SAS scripts" [39]; Elevance "SAS or equivalents" [118]; KPMG [120]; SAS certification an "asset" at National Bank [41]	Keep it as a migration story: ship the SAS-parity mini-project (already P1); recognize PROC SQL / DATA step / macro vocabulary. SAS is ~9–13% of skillenai's job-index postings — real but fading [136]
Hadoop / Hive / MapReduce / Pig	Spark (PySpark) on Databricks + Delta / Iceberg; cloud lakehouse	Big-bank stacks mid-migration — Scotiabank lists Hadoop beside Snowflake and GCP [36]	"Handle, don't master" Spark, as planned; never claim cluster ops
Teradata / DB2 / Oracle DW appliances	BigQuery / Snowflake / Databricks + dbt	Elevance wants "Teradata, SQL" [118]; Loblaw runs Teradata under its analyst stack [123]	Warehouse-agnostic SQL + one modern warehouse in the portfolio; "I migrated X to Y" is a winning interview line
SSIS / Informatica / Talend (ETL suites)	dbt + Airflow (+ managed ELT)	Verified low-frequency but persistent legacy cluster: SSIS 171, Informatica 145 UK postings [133][134]	Recognition-level; describe dbt as "the modern stored-procedure + scheduler layer"
QlikView / Business Objects / SSRS / Crystal	Power BI / Tableau / Looker	QlikView holds 21 UK postings [135] and HSBC lists it beside modern tools [105]; SAP-heavy and Microsoft-heavy estates keep BO/SSRS alive	Recognition-level only — your BI proof stays Power BI + the Tableau mirror
VBA-era Excel reporting	Power Query (M), Python automation, scheduled refresh, Power Automate	VBA is still written into live ads (Manulife [122], Loblaw [123])	Ship the Excel/Power Query workbook as the "automated the manual" story; replace, don't learn VBA
Stored-procedure-mediated ETL + cron	ELT: version-controlled dbt models orchestrated by Airflow	DB-heavy bank shops; TD pairs Power BI with Power Automate work [34]	The repo demonstrates it; "dbt = stored procs that live in Git; Airflow = cron with visibility"
Excel-only dashboards, static PDF/email reporting	Semantic models + interactive dashboards with scheduled refresh	Smaller orgs and legacy teams everywhere	Use the before/after hook: the manual report vs the 30-second refresh (already in the case study)
Untracked notebooks / one-off scripts	Versioned repos, tests, CI, orchestration, monitoring	The anti-pattern senior screens filter for [27] [49]	The repo structure itself is the proof (src package, dbt tests, CI badge)
CSV / Excel as the integration layer	Parquet on object storage + warehouse tables	Legacy shops	Already the design (Parquet staging)

Retiring skill (don't build new)	Modern replacement (build this)	Where it still shows up	Your move
Single-node pandas at any volume	DuckDB / Polars / Spark tiering	—	Documented scale tiers in the README
SPSS / MATLAB- style analysis	Python (statsmodels / scikit-learn) + R	MATLAB accepted as a secondary language at Amazon [1]	One line of awareness; Python stays the headline

How to present the pairings. Keep legacy names out of your headline skills. List them once, honestly, as a migration line on the CV or in interview answers: "translated a SAS-based reporting workflow to Python with parity tests", "migrated stored-procedure logic to versioned dbt models". That framing matches what senior analytics work actually is in 2026 — most mature companies are mid-migration themselves, and the analyst who bridges old and new is worth more than one who only knows the new weather.

The Rigor Standard: What Survives Both Graders

This section is the heavyweight difference. The same repo is read twice — once by a professor who grades whether your claims are *warranted*, once by a manager who grades whether they are *operable* — and Rev 5 designs for both explicitly.

Methodological rigor (the professor reads this). Train/test discipline with dependence-aware splits and preprocessing fitted inside training folds only; a written leakage audit [199]. Backtest the way regulators mean it: Kupiec's unconditional-coverage test and Christoffersen's conditional-coverage test for risk models [194][195]. Both sit under the Basel three-zone framework consolidated as MAR32 (which adds the P&L-attribution leg) [196][197]. For forecasting, rolling-origin evaluation is the standard because random k-fold on time-ordered data is a leakage machine [198]. Every point estimate wears an interval, and probability scores are calibrated before they feed a threshold [208]. Causal claims pass an identification standard — state the assumption in words a skeptic can attack, show the first-stage F for IV, the parallel-trends evidence for DiD, and a placebo test — or label the finding associational [200][201]. And the 2026 governance vocabulary is current: SR 26-2 (US, non-binding, GenAI out of scope) and OSFI E-23 (Canada, effective May 2027 — inventory, lifecycle, explainability-scaled ML) [171][193].

Engineering rigor (the manager reads this). One-command rebuild from a cold clone (uv.lock + seeded generator + make rebuild); data tested like code (dbt tests on schema/ranges/freshness, plus data contracts per dataset); lineage traced raw-to-report with artifact hashes [206]. Documentation reads as *re-executable reasoning*, not narrative.

Statistical communication (both read this). A model card per model in the Gebru/Mitchell sense — intended use, training data and provenance, evaluation with intervals, calibration status, limitations, monitoring pointer [204][205]; a regimes/assumptions register ("trained on 2015–2024 data; not stress-tested for 2008-like tails"); and uncertainty on every stakeholder-facing number — coefficient plots with intervals, fan charts on forecasts, calibration diagrams beside any probability score [203].

Governance artifacts a bank manager recognizes on sight (four files, none requiring bank employment): a MODEL_REGISTRY.md (every model versioned, risk-rated, with review dates, E-23's inventory in miniature), a VALIDATION.md written in second-line voice (challenge your own model harder than you promote it), a MONITORING.md (metrics, thresholds, who gets paged, retrain triggers), and a disciplined CHANGELOG.md. [193][207]

The academic companion artifact. Convert the same project into a professor-legible **methods paper / technical report** (15–25 pages, Quarto/LaTeX, built from the same code): abstract with headline result *and its uncertainty*, data section with sample-construction waterfall, methods with identification assumptions stated as assumptions, results with intervals throughout, robustness with placebos and failure modes, and a reproducibility statement. That document doubles as the portfolio's capstone whitepaper — the professor can referee it, the manager can forward the executive page.

The literature spine, cited locally. Cite at the decision, not in a bibliography dump. Varian anchors the out-of-sample discipline [199]. Kupiec and Christoffersen anchor the backtest and Hyndman & Athanasopoulos anchor rolling-origin evaluation [194][195][198]. The Mixtape and The Effect anchor causal design [200][201]. The ASA statement anchors the p-value policy and Rudin governs model-class choice for high-stakes scoring [202][203]. Datasheets and Model Cards cover the documentation layer [205][204]. SR 26-2 with OSFI E-23 covers the governance layer [171][193].

The anti-pattern list — what gets portfolios rejected on sight by both audiences: overfitting demos (in-sample metrics paraded as skill), p-hacking optics (stars everywhere, no multiplicity handling), uncalibrated probability scores, unexplained black-box usage where stakes are high, **no error analysis**, random splits on time-ordered data, notebook soup as the deliverable, and governance cosplay (files named VALIDATION.md with boilerplate content — worse than none). [209][210][203]

The Flagship Project: NorthLedger

The one-line pitch (use verbatim on résumé and LinkedIn):

"Built NorthLedger, an end-to-end financial intelligence platform that turns 10M+ raw transaction records into executive decisions — Python → BigQuery → dbt → Airflow → Power BI — powering fraud detection (XGBoost, PR-AUC 0.9-class), customer lifetime-value and churn models, and an AI-generated weekly insight memo; shipped as three production-grade dashboards and a documented case study."

The story you tell. A mid-size digital bank has two expensive problems: fraud losses and silent customer churn. Both are buried in the same transaction stream. NorthLedger ingests that stream, models it into a governed warehouse, and serves three audiences: fraud analysts triage alerts with model scores and cost data; the growth team sees who is about to leave and what a saved customer is worth; the executive committee gets a one-page weekly read — including a machine-written narrative memo — of what changed and what to do. Every number in every view traces back to a tested dbt model. That last sentence is what separates a portfolio project from a homework assignment.

Scope — and deliberately what is NOT in scope. In: three analytics modules (Fraud & Risk, Growth & Retention, Executive Intelligence), one warehouse, one orchestration layer, three Power BI reports plus a Tableau Public mirror, a stakeholder Excel/Power Query workbook, an AI insight service, a governance pack (with privacy/regulatory annex), full documentation. Out of v1 (listed as "future work" instead): true streaming (Kafka), deep learning architectures, real bank integrations, mobile apps. Senior engineers cut scope explicitly and say why — the case study will show these decisions.

Architecture.

SOURCES	INGESTION	WAREHOUSE (BigQuery)	SERVING
-----	-----	-----	-----
PaySim / IEEE-CIS transaction data LendingClub (credit) FRED macro API Toronto Open Data Synthetic generator	--> Python extractors (requests, pandas, polars) schema + row checks --> Parquet staging --> GCS / local Airflow DAGs GitHub Actions CI	--> staging (raw, typed) star schema: fact_transactions fact_loans dim_customer (SCD2) dim_account, dim_date, dim_merchant, dim_geography, dim_channel, dim_risk_band marts: fraud / growth / exec + dbt tests + docs + lineage	--> Power BI: Executive Power BI: Fraud Ops Power BI: Growth AI insight layer (LLM weekly memo) text-to-SQL demo Case study + video
ML LAYER: Python notebooks -> model outputs written back as tables (scores, segments, forecasts) -> surfaced in BI			

Data model highlights (the part interviewers dig into). fact_transactions at transaction grain with degenerate dimensions and fraud labels; dim_customer as SCD Type 2 so churn and CLV are measured against the customer's state at decision time; dim_date supporting time intelligence in both dbt and DAX; risk bands and segments as conformed dimensions so the fraud and growth modules share one truth. Every mart model carries dbt tests (not-null, uniqueness, accepted values, relationships) and freshness SLAs; the decision log records why star schema over alternatives at this scale. The interviewer questions you must be able to answer, per the research: the grain of your fact table; how you handle late-arriving and duplicate events; how you would halve the query cost; what your precision/recall trade-off means in dollars; what breaks at 10x volume; and whether the pipeline rebuilds in one command.

Pipeline spec. Daily batch assets (extract → stage → test → transform → publish → notify) with retries and failure alerts; a backfill job; incremental dbt models for the transaction fact (partition by day, merge strategy); a data-quality job running tests and freshness checks that fail loudly; pipeline health surfaced on a monitoring page of the executive report. CI on GitHub Actions: dbt compile + tests on every pull request — that one sentence outclasses most analyst portfolios. Orchestration: **Airflow** is the recommended runner — it appears an order of magnitude more often than Dagster in ads (231 vs 23 UK postings [84][85]) and runs for \$0 locally; Dagster stays as reading knowledge [68]. Pick one and document why; do not run both.

The AI/ML menu (all explainable, all honestly evaluated). Fraud: XGBoost with class-weighted training plus an IsolationForest baseline for unsupervised comparison; reporting focuses on PR-AUC, precision/recall at the operating threshold, and cost saved vs false-positive burden — any fraud professional will ask about the threshold trade-off first

[26]. Churn & CLV: gradient boosting with SHAP explanations driving a retention playbook and a value-at-risk dollar figure. Forecasting: Prophet or ARIMA at multiple grains (portfolio, segment, geography) with a seasonal-naive baseline and rolling-origin backtest, plus a scenario layer (rate shock, loss spike). Every model gets a one-page model card: purpose, data, metrics, limitations, ethics notes. Fraud datasets are synthetic or sampled — say it plainly and explain why the evaluation rigor is the point.

The AI layer — your unfair differentiator. A service that pulls the week's KPI deltas from the executive marts and asks an LLM (a cheap API model; single-digit dollars per month, Section 8) to draft a structured executive memo: what changed, likely drivers, what to watch. Hard guardrails: the model may only narrate numbers handed to it; a validation step checks every figure in the draft against the source table before the memo is "published". Plus a small text-to-SQL demo over the marts with an evaluation harness (you must be able to show its failure rate, not just its demos). As an AI-automation instructor, you can speak to this layer better than almost any competing candidate — and it maps directly to what NVIDIA ("leverage agentic AI") [25] and Amex ("apply emerging AI and GenAI techniques") [45] now ask for.

Credit-risk extension (P1, heavy bank value). Using the LendingClub loan panel: a simplified PD scorecard, vintage/cohort loss curves, and an expected-loss estimate framed as $PD \times LGD \times EAD$. This deliberately speaks the vocabulary that gates senior roles at TD and National Bank (IFRS 9/ECL, RWA, stress testing) [41][46] — with public data and honest simplifications.

The dashboard suite. (1) **Executive Command Center** — KPI cards (revenue, fraud-loss rate, retention, forecast), trend with forecast band, scenario slicer, drillthrough, and the weekly narrative memo embedded as a text panel. (2) **Fraud Operations** — alert queue sorted by risk score, precision/recall dials against the live threshold, cost-saved vs false-positive burden, geographic and merchant heatmaps, model-version field parameter. (3) **Growth & Customers** — RFM/segment matrix, CLV by cohort, churn-risk worklist, retention curves, and the A/B campaign readout with lift and confidence intervals. Design bar: a custom theme, one layout grid, mobile layout pass, and accessibility checks. Publishing rules and the Mac platform path are Section 7.

The heavyweight artifact system (what makes this grade-A for both audiences). Beyond the standard repo, ship: `LEAKAGE.md` (every feature's availability timestamp, split logic in code, a CI leakage test), a `BACKTEST_REPORT.md` per model (Kupiec/Christoffersen statistics with p-values and effect magnitudes, Basel three-zone verdict, decision rule on amber/red) [194][195][196], a `VALIDATION.md` in second-line voice, `MONITORING.md`, `MODEL_REGISTRY.md`, an error-analysis appendix on every model card, the metric-certification spec, and the 15–25 page methods paper generated from the same sources. On the operations side a manager screens for: `COST.md` (bytes scanned per refresh, monthly run cost, what you'd cut at 10× scale), `RUNBOOK.md` (what breaks, how you know, who you page, rollback), a simulated `INCIDENT_POSTMORTEM.md` (schema-drift or late-data incident in a director-facing format) [211][212], and a one-page `REQUIREMENTS.md` written as if to a finance stakeholder.

Repository structure (what a reviewer sees first).

```
/northledger
├─ README.md           <- the case study: problem > data > architecture > results > impact > how to run
├─ /architecture       <- ERD, pipeline diagram, decision log (ADRs)
├─ /pipelines          <- Airflow DAGs, extractors, dbt project (models/, tests/, docs/)
├─ /analysis           <- showcase SQL (20 queries), experiment readout, ad-hoc notebooks
├─ /models             <- training notebooks, model cards, evaluation reports
├─ /ai                 <- insight memo service, text-to-SQL demo, guardrails + eval
├─ /bi                 <- Power BI theme JSON, DAX snippets, report screenshots, Tableau mirror, publish
links
├─ /excel              <- stakeholder workbook (pivots, Power Query, scenario model)
├─ /docs               <- data dictionary, KPI definitions, lineage, reconciliation, runbook
└─ /assets             <- screenshots, demo video link, case-study PDF
```

Governance pack (the banking differentiator). Ship the exact artifacts bank postings name [42][45]: data dictionary · lineage diagram (dbt docs) · data-quality test suite [36]. Plus the audit-grade layer: reconciliation report · model cards · decision log · and a **metric-certification spec** — one page per executive KPI (formula, grain, owner, freshness SLA, change policy), the artifact behind "source-of-truth datasets" language in senior postings [1][16]. For EU or regulated targets, add the **privacy/regulatory annex**: a DPIA-style note, PII handling, and a risk-classification table mapped to EU AI Act categories — on the 2026 calendar: transparency duties and GPAI enforcement are live (Aug 2, 2026) and Annex III high-risk obligations were deferred to Dec 2, 2027 by the Digital Omnibus [90][91][216]. Most portfolios have none of these;

every senior bank posting asks for them. **Finance & economics bolt-ons (thin modules on the same flagship, ≈25–40 hours, no new datasets)**. An **AML layer** adds typology rules, alert triage and a case narrative to Module A (≈8–12 h) [137][139]. The **ECL staging + stress-test** upgrade adds SICR triggers and scenario-weighted PD (≈9–12 h) [141][151]. A **regulatory annex v2** maps BCBS 239 to dbt controls and names scorecard methods (WoE/IV, Gini/KS) (≈4–7 h) [152][144]. For the finance-industry track, a **markets spine + VaR-lite engine** adds a security master, corporate actions, and a Kupiec-backtested VaR with model card (≈50–70 h) [159][160][169]. For the econ track, an **econ toolchain kit** adds a Stata do-file, an R/Quarto mirror, and the panel/DiD/VAR upgrade (≈11–19 h) [148][189][187]. Optional: an **insurance micro-module** (exposure aggregation + return-period table) [172][173].

Power BI on a Mac: Platform and Publishing Rules

The honest constraint: **there is still no native Power BI Desktop for macOS, and no announced plan for one** — Microsoft's own installation documentation requires Windows [61]. But the 2026 reality gives a Mac user three workable paths, and the recommended combination costs \$0 to start.

Path	Cost	What works / what breaks
Browser authoring (app.powerbi.com)	\$0	Get Data, Power Query (M), semantic-model editing and DAX now work in the browser [62]. Breaks: no local <code>.pbix</code> , no offline work, some desktop-only connectors, and large-model conversion failures silently drop you to Viewing mode [62]
Local Windows VM (recommended)	\$0 with VMware Fusion (free for personal use) or ~\$99.99/yr Parallels or free UTM [73]	Full Power BI Desktop in a Windows 11 Arm VM (needs the 2025-09 ARM update KB5065789 [61]; Power BI Desktop runs under x64 emulation with reported rendering/refresh instability — keep browser authoring as the escape hatch on bad days); sizing that works: 4 vCPU / 6–8 GB RAM / 80–120 GB disk [73]. Keep <code>.pbix</code> files in a shared folder so macOS and Windows see the same project
Cloud PC (Windows 365 / AVD)	\$31–66/mo after trial [74]	Desktop "fully supported" on Windows 365/AVD [61]; only worth it if you want zero local setup — bills whether you use it or not

Licensing — where Mac users get burned. A Free Power BI license can build in Desktop and **Publish to web** from My Workspace (that is what your public portfolio dashboards will use) [57]. What Free cannot do is **scheduled refresh** — that needs Pro (\$14/user/mo, 8 refreshes/day) or PPU (\$24/user/mo, 48/day) [72]. Plan: build the whole project on Free; buy **Pro for the 2–3 months when you need scheduled refresh on camera** for the demo video, then cancel. Run the **Fabric 60-day trial** during the data-engineering phase to record Dataflows/notebooks/pipeline evidence before it expires [63] — Fabric is now written into live analyst stacks (Allstate's data-engineering req is Microsoft Fabric [128]; AIB's ad literally says "SQL, Python and Fabric" [109]). The EU bars to copy: BNP Paribas's Power BI specialist ad asks for Desktop/Service/Dataflows, DAX, Power Query (M), plus release management and documentation [107].

Publishing rules that protect you. Publish-to-web makes a report public with no authentication — Microsoft explicitly warns that anyone can access the underlying data in the model, even if the report does not display it [57]. Therefore: publish only reports built on public data (or fully synthetic data), and keep everything else in the repo as screenshots and PDFs. The winning pattern: 2–3 public Publish-to-web links as the "one-click" experience [27], with the GitHub repo as the depth layer.

The Data & Stack Playbook

The whole platform runs for **\$0/month**; you may voluntarily spend \$14–24 in the scheduled-refresh months. This is the stack, verified against 2026 pricing.

Layer	Pick	Cost
Authoring Power BI	Windows 11 Arm VM (VMware Fusion, free for personal use) + Power BI Desktop	\$0
Public dashboard delivery	Publish to web from My Workspace (Free license qualifies) + repo screenshots	\$0
Local core engine	DuckDB + Parquet + Polars — reads tens of GB on a laptop SSD; where ~70% of the engineering happens	\$0
Cloud warehouse	BigQuery sandbox — permanent free tier: 1 TiB queries + 10 GiB storage/month, no card [65]; Databricks Free Edition for the Spark/lakehouse story [66]	\$0
Transform + orchestrate	dbt Core + Airflow locally [84]; GitHub Actions as the free scheduler (2,000 min/month on private repos, unlimited on public)	\$0
ML	scikit-learn, XGBoost/LightGBM, Prophet, PyOD, SHAP, MLflow	\$0
LLM narratives	GPT-4o mini-class API (≈\$3/month for a demo workload; GPT-5 mini ≈\$8.50) [71]; Ollama locally as the \$0 offline variant	\$0–\$10/mo
Scheduled refresh + Copilot (optional)	Power BI Pro \$14/user/mo [72]	\$0 or \$14–24

Cloud warehouse details and the traps. BigQuery sandbox tables **auto-expire after 60 days** — keep raw data as local Parquet and make the warehouse rebuildable in one command [65]. Snowflake offers \$400 credits / 30 days (region locked at signup) — use it as a two-week "enterprise warehouse" sprint, not a home base. Databricks Free Edition is forever-free but serverless-only, quota-limited and **non-commercial** — the realistic way to demo Spark/Delta/MLflow [66]. AWS's 6-month new-account plan closes when credits run out — an S3 + Athena demo, never a system of record. MotherDuck Lite (DuckDB in the cloud) is free at 10 GB / 10 CU-hours if you want a cloud DuckDB story [67]. Microsoft Fabric is spreading through enterprise stacks [128][109] — the Section 7 trial covers that keyword too.

Orchestration — pick exactly one, and the v2 evidence says Airflow. Airflow appears in ads roughly ten times as often as Dagster (231 vs 23 UK postings [84][85]) and runs \$0 locally; Dagster stays as reading knowledge (asset-based, dbt-native [68]); Prefect is fine if you truly never care about keyword screens. GitHub Actions on a daily/weekly cron remains the \$0 scheduler that runs it all. (dbt itself is right but region-shaped — strongest in analytics-engineering and DE-flavored postings [98][95].)

Streaming — one weekend, maximum. A local Kafka (KRaft) or Redpanda micro-batch demo is enough to prove event-time vs processing-time and windowed aggregates; a 24/7 streaming stack on a laptop adds no hiring signal and lots of breakage. Frame it as an extension.

Datasets (sizes measured Sept 2026; pick three, not seven). Selection rule: one volume dataset + one relational dataset + one local Toronto dataset, all serving the one narrative.

Dataset	Size / shape	License notes	Proves
PaySim (mobile-money fraud)	6.3M rows / ≈493 MB	Kaggle, synthetic — cite the authors	fraud labels, imbalanced classification
IEEE-CIS Fraud Detection	≈590 MB	Kaggle competition rules	realistic fraud feature engineering
LendingClub loans 2007–2018	≈1.5–2.2 GB	public releases; verify license before commercial use	credit risk, vintage/survival analysis
NYC TLC trip records	105 MB (yellow 2019-01) → 451 MB (fhvhv month); full corpus 60–100 GB [69]	open data	volume processing, partitioning, geospatial
NOAA GSOD (BigQuery public)	115M rows, 1929→present	public, hosted free	cheap cloud ingress, partitioned scans
NYC 311	22.5M+ rows	public	SLA analytics, text categorisation

Dataset	Size / shape	License notes	Proves
Toronto Open Data (TTC, delays, traffic)	varies	open data [70]	local market relevance

The flagship uses: PaySim + a **self-built synthetic transaction generator** (scale layer to 10M+ rows — building the generator is itself a skill signal), LendingClub (relational/credit), FRED macro series (exec scenario layer), and Toronto Open Data as a P1 enrichment.

The AI layer costs pennies. ML runs in Python jobs (never trapped inside a BI visual) and writes scored tables into the warehouse for Power BI to read. LLM narrative generation for the executive memo runs on the batch API: a demo workload of $\approx 10\text{M}$ input + 3M output tokens per month costs about \$3 on GPT-4o mini-class models and $\approx \$8.50$ on GPT-5 mini-class [71] — less with prompt caching and a weekly cadence. Ollama on Apple silicon is the free, offline alternative and a nice "private/air-gapped variant" line in the README. Ship embeddings/chat-with-data only with an evaluation set; never fine-tune a foundation model for this project.

Alternate Flagship Blueprints

If the financial angle does not fit the target set you settle on, these swap in with the same architecture, roadmap and packaging system — only the domain layer changes. Pick ONE and go deep; do not build two flagships.

Alt A — StreamPower: Media & Streaming Engagement Analytics. One-line: model a streaming service's engagement, retention and content performance on 25M+ ratings plus a simulated event stream. Data: MovieLens 25M + synthetic activity events + content metadata. Shines for: Meta/Netflix/Spotify-class product-analytics and experimentation roles — leans hard into metric design (engagement, retention curves, cohort LTV) and A/B experiments. Swap the fraud module for a recommendation-baseline analysis and content-investment ROI.

Alt B — MapleMacro: Housing & Macro Intelligence for Canada. One-line: an executive intelligence platform on Canadian housing, rates and affordability with forecasting and scenario analysis. Data: StatCan, CMHC, Toronto open data [70], FRED. Shines for: Canadian bank strategy/marketing roles and public-sector analytics — and it weaponizes the MSc Applied Economics; few data candidates can defend causal framing and macro overlays credibly. Weakest fit for Big Tech.

Alt C — CartPulse: E-commerce Growth Analytics. One-line: revenue operations, customer lifetime value and marketing incrementality for an online retailer. Data: Olist (100k+ real orders, CC BY-NC-SA 4.0 — non-commercial, say so) + a synthetic event layer for scale. Shines for: retail/Amazon-style roles; the classic "safe" portfolio topic — which is also its weakness (crowded). Pick it only if retail is your target.

The Open Data Universe: Verified Public Datasets by Domain

The portfolio's raw material problem is solved by public data — but only if the datasets are real, licensed, and downloadable today. The catalogue below is the verified core (each entry was fetched during this engagement); the full **178-dataset master catalogue** — 70 North America, 60 global, 48 Europe — ships as `NorthLedger-open-data-catalog.csv`, one row per dataset with URL, scale, licence and verification status.

Domain	Flagship-verified datasets (access + scale)	Best use
Crime & justice (US)	Chicago Crimes — 8,642,798 rows, live-counted [218] · NYPD Complaint Data Historic — 10,071,507 rows [219] · FBI Crime Data Explorer (NIBRS/UCR) [217] · LA/Seattle/SF police incidents · Eviction Lab	Big-row pipeline demo; geo-temporal analysis; fairness-aware modeling with documented caveats
Transportation	NYC TLC trip records — monthly Parquet, 473 MB for FHV 2024-01 [69] · BTS aviation · GTFS transit feeds · OpenFlights routes	The blueprint's streaming/micro-batch demo; scheduling and network analytics
Cycling & micro-mobility (NL/EU)	CBS mobility tables — 970,200 rows measured [222] · CBS bicycle-network length — 20,328 rows [222] · NDW national bicycle-count network (live feeds) [224] · Fiets TelWeek [233] · RDW registry — 16.9M registered vehicles [223] · Amsterdam open data [225] · TfL cycle-hire journeys, the UK counterpart [221]	The Netherlands cycling/e-mobility track; city-scale sensor + survey fusion
Energy & climate	EIA API + bulk [229] · NOAA/GCP climate corpora · NASA Earthdata · ERA5 · Copernicus Sentinel (satellite)	Time-series at scale; climate-risk overlays for finance use cases
Health	CMS · CDC WONDER · NHANES · MIMIC-IV (credentialed)	Careful: credentialing + privacy notes; health-economics projects
Finance & economics	FRED [232] · SEC EDGAR full-text + financial-statement datasets, ~124 MB of XBRL facts per quarter [228] · FEC filings · US Treasury · Statistics Canada · ECB Data Portal	NorthLedger macro overlay; financial-text analytics; econometrics modules
Housing & population	US Census · HUD · Statistics Canada census · CMHC (Canada)	MapleMacro alternate; housing/affordability analytics
Government & portals	data.gov · data.europa.eu — ~1.6–1.7M dataset listings [226] · Eurostat APIs [227] · data.police.uk [220] · Toronto open data · EU member-state portals	Discovery layer for alternates; metadata + linkage projects
Text / web corpora	GDELT · Wikipedia/Wikidata dumps · common Crawl · Project Gutenberg	NLP pipelines; media-trend analysis
Science & open indexes	Zenodo [231] · Hugging Face datasets [230] · Kaggle · AWS Open Data Registry · Google BigQuery public datasets	Search layer; benchmark baselines against known public results

Selection rules for alternates. Favor datasets with: a working bulk endpoint (not a broken portal), a clear licence, a size that exercises partitioning (>100k rows at minimum; >1M preferred), and a domain you can speak about for 30 minutes. The two you named are both in here and both live-verified: US criminal-record data at city and federal scale [218][219] [217]. Dutch cycling and e-mobility data is covered from three directions — counting networks, the vehicle registry, and the mobility tables [224][223][222]. Every row in the CSV carries its verification status, so nothing ships that was not probed.

The 6-Month Roadmap

Ten to fifteen hours a week gets this done in six months with room for life. The working rhythm: three weeknights (2h each) plus one weekend block (4–6h). The research scoping for a build like this is 80–150 hours (of which roughly 30–40% is data plumbing/quality, 20% modeling, 15% BI/DAX, 25% documentation and storytelling); the full six-month load at 10–15 hours/week is ~260–390 available hours, the remainder going to PL-300 study (~80–120 h, Months 3–5), the skill bolt-ons (~30–50 h if taken), monthly public artifacts, and Month-6 application materials. Plan the whole budget, not just the build — and plan for the plumbing, which is where solo builders quit. One rule: every month ends with something PUBLIC — a commit history, a short post, a dashboard screenshot. Applications start in Month 6, not after "everything is perfect."

Month	Focus	You finish the month with	Gate to pass
M1	Foundations + data	SQL refresher done (window functions, CTEs); Python/polars warm; repo skeleton live; BigQuery + Kaggle + FRED + Power BI accounts; KPI dictionary v1; first raw data landed and profiled	Repo public with README + sustained commit history
M2	Data foundation	Ingestion jobs + Airflow running daily; BigQuery staging + star schema v1 in dbt; data-quality tests; dbt docs published; architecture diagram v1	A green daily pipeline anyone can re-run in one command
M3	Analytics depth + first BI	All three marts complete; 20-query showcase file; Executive report v1 published; case-study skeleton; PL-300 study begins	First dashboard link shareable
M4	ML/AI models	Fraud, churn/CLV and forecasting models trained, evaluated, written back to the warehouse; model cards; evaluation notebook; PL-300 booked or passed	Model metrics defensible under questioning
M5	AI layer, experiments, polish	LLM insight memo live with guardrails; A/B campaign readout with power calculation + peeking note + a caught failed guardrail; credit-risk extension; the rigor pack (leakage audit, backtest reports with Kupiec/Christoffersen statistics, validation memo in second-line voice, model registry + monitoring plan) and the methods paper; all three dashboards polished + Excel workbook and Tableau mirror shipped; monitoring page	Full stack demo-able end-to-end in 10 minutes
M6	Packaging + applications	Case study PDF; 90-second video; LinkedIn series (8 posts); portfolio website (single-file, dual-door: "Hire me" + "Book a data audit", Section 14) ; résumé variants; interview pack; applications out	Applying with a live, verifiable portfolio AND a client-facing door

Priority discipline (protect it ruthlessly). P0 must exist before applications go out: the pipeline, the star schema, the three marts, one defensible model per module, three dashboards, the **Excel/Power Query workbook and the Tableau mirror**, the governance pack, the case study, the video, the README. P1 adds the AI layer, the A/B experiment, causal-inference analysis, the SAS-parity mini-project, the RLS demo, the Git/Agile sprint-report skin, and — for EU or regulated targets — the privacy annex and an MLflow-tracked model run. P2 is everything else. Chasing finance-industry or economic roles? Fold the Rev-4 bolt-ons (AML layer, ECL/stress upgrade, regulatory annex v2, econ toolchain kit) into the P1 set — they are extensions of existing modules, not new workstreams (Section 6). If Month 5 arrives and a P1 item is not done, ship without it — a finished project beats a perfect half-built one, and the case study can list it under "future work" honestly.

How to Present It So You Stand Out

Read by a manager — the five-minute path and what gets you rejected. A technical reviewer spends ~30 seconds on the first pass and ~3–5 minutes total: repo landing page (README with a one-line problem, a quantified headline result, and a live/runnable artifact), then a runnable check (structure, orchestration, tests, CI, pinned environment), then one deeper click for judgment under constraint. Instant rejections: tutorial datasets with unchanged framing, metrics without a baseline, no leakage check, raw notebook dumps, dead dashboard links, overengineering a simple question, and unedited AI boilerplate [209][210]. The seniority signals that separate "candidate" from "colleague": cost-awareness (a `COST.md`), monitoring with thresholds and rollback (`RUNBOOK.md`), a simulated incident postmortem (`INCIDENT_POSTMORTEM.md`), a BRD-lite requirements doc, decision logs, and "what I'd change" sections [211][212]. The "forward to my executives" test is passed by exactly three artifacts: a decision-first one-page executive summary (ask → status → recommendation) [214][215], a public clickable artifact at a stable URL [213], and a requirements doc written to a stakeholder. The same artifact reads twice: management framing opens with decision and dollars (method in the appendix), academic framing opens with method and identification (decision in the conclusion) — ship both (executive page on top, methods paper deeper).

The attention economics are brutal; design for them. Reviewers give a portfolio roughly 30 seconds to 3 minutes [49]; the decision to keep reading happens in the first 30 seconds [49]; a repo with a tool-named title gets about four seconds [51]. About 95% of portfolios fail the one test that matters — connecting the analysis to a decision or outcome [48]. Three strong projects beat ten mediocre ones [50], and the reviewer will not clone your repo or read a 15-page notebook [49]. One honest counter-signal: some employers report that portfolios rarely land the job by themselves — treat yours as the ammunition you fire in interviews rather than a passive screen-passer [31]. Everything below is built around those facts.

The two frameworks to write with. SCQA (Situation → Complication → Question → Answer) for problem framing, and the Minto Pyramid (answer first, then grouped support, then evidence) because it matches how executives process information [55]. The hiring-manager-authored interview variant is **CRAIG**: Context, Role, Actions, Impact, Growth [48]. Use SCQA/Minto in the written case study; use CRAIG for how you talk about the project in interviews. Hiring managers also score portfolios on a five-dimension rubric — problem framing, data realism, evaluation rigor, deployment thinking, and communication — and the anatomy below is built to satisfy all five [52].

The README is the product. Keep it under ~500 words; full detail lives in linked notebooks and the case-study PDF. Anatomy that survives the 30-second skim: (1) headline with the business problem, the finding and the number in one line; (2) TL;DR box, ≤80 words, recommendation first; (3) situation & complication — stakes and tension; (4) data & quality — sources, volume, what was broken; (5) approach — plain English first, the 2–3 decisions you would defend; (6) findings — maximum three, each with one chart and one "so what"; (7) recommendation & quantified impact — action, owner, cost, expected value; (8) limitations & next steps — what production would need; (9) repository guide. A README that opens with the stack has already lost the room [51].

Quantify like an operator, not a student. State baselines explicitly: "manual reporting takes ~9 analyst-hours/week; the automated model refreshes in 30 seconds" [56]. Put model claims in business units: "flagging the top 5% of transactions by risk captures 62% of fraud value at a false-positive cost of \$0.90 per reviewed alert." Report precision/recall with costs — "99.8% accuracy" on imbalanced data signals you do not understand the problem [50]. And document the cleaning: roughly 60–70% of real work is data cleaning, and saying so is a credibility signal when placed after the business hook [47] [56].

Video, publishing, and the one-click test. Record a 90-second Loom: 0–10s problem and stakes, 10–30s data and method, 30–60s the two findings on screen, 60–90s recommendation and what you would do next. It solves the "reviewer will not clone your repo" problem [49] and shows communication, which is the skill every loop tests. For the dashboards: publish only public-data reports via Power BI Publish-to-web (remember: anyone can access the underlying model data, so public data only) [57], and mirror one dashboard on Tableau Public for the US firms that name Tableau [43][58]. Maven Showcase is a free, purpose-built venue that recruiters browse — cross-post there [58]. The goal: one URL and the reviewer is looking at your work — they will not clone the repo and they will not dig [49].

The personal website is optional polish for hiring — but load-bearing for clients. For employers the evidence still holds: a survey of 60+ hiring managers found most will click a personal site but it does not change their decision, and unmaintained sites backfire [53] — for them the GitHub profile README stays the landing layer [53]. For *business clients* the website is where trust converts: they are not screening a candidate, they are buying an outcome, and they need a

page that says what they get, what it costs to start, and how to book. Rev 6 therefore ships one lean single-file site (Section 14) with two doors on one page — "Hire me" → the case study and live dashboards; "Book a data audit" → the diagnostic offer. Keep it shallow and maintained; a neglected client site reads worse than none.

LinkedIn build series. Post one short public update per milestone (problem framing → data sourcing → cleaning war stories → findings → dashboard → recommendation), each linking the artifact [60]. It creates pre-interview familiarity, doubles as narrative rehearsal, and recruiters search it. By application time, your project has a trail — not a single link.

Résumé rules. The first scan is 6–10 seconds [56]. List three to four projects maximum, each one line in problem → action → quantified result form with a link [50]. Mirror the postings' exact vocabulary (the senior-analyst cluster: SQL, Python, data modeling, ETL, dashboarding, Power BI/Tableau, KPI definition, stakeholder management, experimentation, data quality/governance [42][45]). Keyword-literalness matters — the exact strings **Excel, Tableau, Git, Agile/Scrum/Jira, Power BI, Databricks** are what ATS screens match, and a concept without the string often scores as absent [96][75]. Keep resume and portfolio speaking the same language — candidates who fix that mismatch report their screens changing from "walk me through your background" to actual project questions [49].

Examples that show the bar. Shirisha-S11's supply-chain analytics repo opens with a business question, states its stack up front (SQL + BigQuery ML + Looker Studio — no Python at all), and ships a live dashboard link; honest scoping beats false breadth [59]. Harrison Jansma's site leads with three exemplar projects rather than everything he has ever done. The Maven Showcase hosts purpose-built project portfolios that hiring managers browse directly [58]. None of them are elaborate; they are clear.

Certifications That Actually Move the Needle

Certs clear ATS filters; the portfolio wins interviews. And 72% of hiring managers say a portfolio matters more than a certification [56]. With that framing, here is the ranked verdict:

Certification	Cost	Time	Verdict
Microsoft PL-300 (Power BI Data Analyst Associate)	\$165	80–120 h	Do this one. Highest-ROI single credential for this pivot: it maps one-to-one to a screened job title, appears as a preferred qualification in postings, renews free annually [54] — and TD's live Power BI posting lists "Power BI certification" as an explicit asset [34]. Official credential page: [64]
Microsoft DP-600 (Fabric Analytics Engineer)	\$165	~100 h	Second pick — take it after PL-300 if you are targeting banks on the Microsoft stack (CIBC/TD Azure + Databricks shops) [40]
One cloud cert (Azure or AWS)	~\$100–165	2–4 months	Just-in-time add-on: AWS certification appears as a requirement in 19.7% of data-scientist postings and Azure co-occurs in roughly a quarter of UK Power BI ads [29][88] — take ONE, matching the stack you target
Google Advanced Data Analytics	~\$245–295	5–6 months	Skip. Your MSc Applied Economics out-signals it and the certificate itself promises nothing; the portfolio does the heavy lifting in this space [54]
AWS Certified Data Engineer – Associate	\$150	2–4 months	Only for a data-engineering-leaning pivot; practical experience generally outweighs it
Databricks Certified Data Analyst Associate	\$200	6–10 weeks	Relevant for lakehouse shops (CIBC "Azure Databricks is a must" [40]); optional

Banking-sector extras worth knowing: National Bank lists **SAS certification** and **CFA/FRM progress** as assets for its senior credit-risk data science role [41] — treat these as optional context if you go deep on credit risk, not near-term requirements. Recommended sequence: PL-300 study during Months 3–4, exam in Month 4–5 (it also forces DAX depth you need anyway), DP-600 only if bank targets demand it, then at most one cloud cert for the stack you target. Vendor badges (SnowPro, dbt, Tableau) show up in marketing more than in posting evidence — treat them as optional and skip unless a target shop names them [99]. For finance tracks: CFA / FRM / PRM appear only as tiebreakers ("asset", "a plus") in the finance postings reviewed [167][159] — no study time now; revisit only if three or more target postings name it.

Risks, Traps, and the Honesty Rules

- **Synthetic and sampled data.** Some fraud datasets are synthetic; some panels are samples. Never pretend otherwise. State it in the README and explain why evaluation rigor, not data authenticity, is the skill being demonstrated. Interviewers respect the honesty and punish the oversell.
- **Scope creep is the project killer.** The "NOT in scope" list is a feature. Freeze scope at Month 2; park every new idea in `/docs/future-work.md`. The solo-build failure modes are documented in the research reports: five domains in one project, a model zoo instead of one evaluated model, and streaming stacks nobody asked for.
- **Plan for the plumbing.** 30–40% of the effort is data plumbing and quality. Solo builders underestimate it by 2–3x and quit there. Budget it explicitly.
- **Design for the skim.** Reviewer attention: 30 seconds to decide, 3 minutes maximum [49]; README under ~500 words [50]; no digging required [49]. If a reviewer has to work, they leave.
- **Reproducibility is non-negotiable.** A fresh clone plus one command must rebuild everything. Every number in the case study must be regenerable by running the pipeline. No hand-typed numbers — if a reviewer cannot reproduce it, it does not count.
- **Dashboard aesthetics are a signal.** A default-theme report reads as junior. Budget the Month-5 polish pass: one theme, one grid, consistent number formats, accessible colors.
- **Avoid the exhausted datasets.** Titanic, Iris and stock-price dashboards have been seen a thousand times and read as coursework [50]. Domain-fit beats topic comfort.
- **Weight the market correctly.** Excel and Tableau are the highest-frequency analyst mentions; ML depth and causal inference are DS-track differentiators — do not let DS-depth work crowd out the analyst table stakes, and do not lead with ML for analyst roles [75][76][95].
- **Documentation debt compounds.** The decision log (why star schema, why this threshold, why this forecast method) is written during the build, not reconstructed later. It doubles as interview preparation.
- **Decide Power BI's Mac path in Month 1** (Section 7), not Month 3. Record the Fabric-trial data-engineering demo before the 60-day window closes [63].
- **Don't build in silence.** The monthly public post is part of the plan, not optional polish — it creates the paper trail that recruiters and hiring managers actually find [60].

From Portfolio to Paid Work: The Client-Facing Offer (NorthLedger Insights)

Hiring managers and professors are two doors; the third is the market **selling to small and mid-size businesses** — the shops drowning in exports, spreadsheets, and five disconnected tools. NorthLedger is the proof; **NorthLedger Insights** is the offer. The point of this section is not theory: it is the exact service ladder, the messy-data-to-live-insight loop you demonstrate on public data, and the packaging that turns the portfolio into revenue. Everything below is honest — the project runs on public/synthetic data (stated plainly), and the skill being sold is the *engineering*, not the data.

Who this is for — and what you are actually selling

The buyer is an owner or operations lead at a service business, retailer, agency, clinic, or growing SMB. Their data is messy: POS/e-commerce exports, booking-system CSVs, ad-platform screenshots in spreadsheets, a CRM somebody half-filled in, and a "monthly report" that is one person re-typing numbers. Decisions run on instinct because reporting is a chore. They do not need a data scientist on payroll — they need their data to **clean itself, model itself against their goals, and tell them every morning what changed and what to do about it**. That is the offer. You are selling the pipeline and the daily insight layer — the thing NorthLedger already is — not hours of analysis.

The messy-data → daily-insight loop (what the website shows live)

This is the story the site tells with a working demo, in the buyer's language. Each stage maps to a NorthLedger module, so the demo is real engineering, not a mockup.

Stage	What the client hands you (messy)	What your automation does	What they see every morning
1. Collect	CSV exports, spreadsheet copies, API pulls, emailed reports	Scheduled Airflow ingestion (the pipeline that already runs daily in the flagship)	Nothing to touch — it just arrives
2. Clean	Duplicates, inconsistent formats, missing fields, mixed currencies/dates, typo'd categories	dbt staging + tests: dedupe, standardize, typed casts, rejected-row quarantine, freshness checks	A small "data health" line: X rows in, Y fixed, Z rejected and why
3. Model	Rows with no meaning attached	Star schema (one truth), KPI layer built from their objectives (sales, margin, churn, no-shows — whatever pays the bills)	Numbers that agree with each other, not five different spreadsheets
4. Insight	A wall of numbers	The AI weekly/daily memo over the marts — number-validated, guardrailed (Section 6): what changed, the likely driver, what to watch	A short, plain-language briefing they actually read
5. Live view	Static reports	Dashboard that refreshes on schedule (the executive/growth dashboards)	Live trends, forecasts with bands, alerts on movement
6. Story + prediction	Hindsight only	Rolling-origin backtested forecasts + scenario layer (rate shock, demand spike) + the narrative tying trend → cause → next step	"Here is what happens next if this continues" — with stated uncertainty

The service ladder (productized, not hourly)

Three rungs that step a client from skepticism to retainer. Prices are honest market anchors for a junior-to-mid solo consultant in Toronto/remote — anchor to value, stay flexible, and always sell the *outcome* first.

Offer	What it is	Time to deliver	Price anchor (CAD)	The client gets
1. Data Health Audit (entry offer)	Fixed-scope diagnosis of their current data: what exists, what is broken (duplicates, gaps, format rot), what is automatable, and the top 3 wins — delivered as a scored report + a 30-minute call	1–2 weeks	~\$500–1,500 fixed	Clarity: a written verdict and a prioritized fix list, whether or not they continue
2. Automated Insights Build (the flagship service)	The full pipeline: connect sources → clean → model against their objectives → live dashboard → daily LLM insight memo → forecast layer. Includes a handover runbook and 2 training sessions	4–8 weeks	~\$4,000–10,000 setup (scales with source count and complexity)	An end-to-end system that runs itself and pays for itself in analyst-hours saved

Offer	What it is	Time to deliver	Price anchor (CAD)	The client gets
3. Insights Retainer (operate plan)	Pipeline monitoring and maintenance, monthly insight summary, one ad-hoc analysis, model/forecast refreshes, small enhancements — the "your data person on call" tier	ongoing	~\$800–2,500/month	Someone making sure it keeps working and keeps telling them what to do

Rules that protect you: **fixed scope on the build** (scope creep kills solo consultants — the flagship's own "NOT in scope" discipline applies to clients too); **automation is the product, not your hours** (the more you automate, the higher your effective rate); **start every engagement with the audit** (it de-risks both sides and surfaces the real objective); and **never promise a number you cannot regenerate** — the same honesty rule as the portfolio, now a contract term.

The one-page capstone: build it free, show it live

The offer needs one runnable proof a prospective client can click without a sales call. Build a **public-data capstone** in Month 5–6 that mirrors the loop end-to-end on \$0 infrastructure: pick a relatable public dataset (Toronto Open Data for a local-services story, StatCan/FRED for an economics story), run it through the same clean → model → dashboard → daily-memo loop, generate a week of dated briefings, and embed the live dashboard plus the memo archive on the website. Publish the dashboard via Power BI Publish-to-web (public data only — Microsoft's warning stands [57]) and keep the pipeline public. This is the artifact that converts "can he really do this?" into "how much to do it to my data?" — and it doubles as interview ammunition, because running one loop on a second domain proves the architecture generalizes.

Selling honestly (the rules are the same as the portfolio)

Disclose that the demo runs on public/synthetic data — the craft is the pipeline. Never sell accuracy theater on imbalanced problems. Put the forecast's uncertainty in the memo, in words. And treat client data as confidential from the first call — that instinct, stated plainly on the website, is itself a selling point for regulated-adjacent SMBs (clinics, finance-adjacent services). The client-facing website (Section 12) carries this section's table, the audit's booking path, the live capstone, and the same case study that closes employers.

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